

## Grammar - Results



### Attempt 1 of 1

Written Jan 27, 2026 10:18 AM - Jan 27, 2026 11:16 AM

Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

#### Question 1

1 / 1 point

Which statement is true?

- ☐ Every context-free language is regular
- ☐ Context-sensitive languages are undecidable
- ☐ Type-0 languages are always finite
- ☒ Every regular language is context-free

#### Question 2

1 / 1 point

Which grammar type allows length-decreasing productions?

- ☐ Type-3 only
- ☐ Type-2 only

- ☐ Type-1 only
- ✓ ☒ Type-0

### Question 3

1 / 1 point

which language is CFG but not regular

- ✓ ☒  $\{ a^n b^n \mid n \geq 0 \}$
- ☐  $\{ a^n \mid n \geq 0 \}$
- ☐  $\{ (ab)^n \mid n \geq 0 \}$
- ☐  $\{ \epsilon \}$

### Question 4

1 / 1 point

which language is context sensitive but not context free?

- ☐  $\{ a^n b^n \mid n \geq 0 \}$
- ✓ ☒  $\{ a^n b^n c^n \mid n \geq 0 \}$
- ☐  $\{ (ab)^n \mid n \geq 0 \}$
- ☐  $\{ a^n \mid n \geq 0 \}$

### Question 5

1 / 1 point

Which grammar type corresponds exactly to Turing Machines?

- ☐ Type-3

- ☐ Type-2
- ☐ Type-1
- ✓ ☒ Type-0

### Question 6

1 / 1 point

Which generate the language of matching parentheses?

- ☐  $S \rightarrow (S) \mid \epsilon$
- ☐  $S \rightarrow (S( \mid \epsilon$
- ☐  $S \rightarrow ()S \mid \epsilon$
- ✓ ☒  $S \rightarrow SS \mid (S) \mid \epsilon$

### Question 7

1 / 1 point

How many parse trees does  $()()$  have for the grammar

$S \rightarrow SS \mid (S) \mid \epsilon$ ?

- ☐ 1
- ✓ ☒ 2
- ☐ 3
- ☐ 0

### Question 8

1 / 1 point

How many parse trees does  $()()$  have for the grammar

$S \rightarrow (S)S \mid \epsilon$  ?

- ✓ ☒ 1  
☐ 2  
☐ 3  
☐ 0

### Question 9

1 / 1 point

which of the following define the language

$$L = \{a^n b^m \mid n \geq m \geq 0\}$$

- ☐  $S \rightarrow aSb \mid aaSb \mid \epsilon$   
☐  $S \rightarrow aS \mid bS \mid \epsilon$   
☐  $S \rightarrow aSb \mid \epsilon$   
✓ ☒  $S \rightarrow aSb \mid aS \mid \epsilon$

### Question 10

1 / 1 point

which of the following generate the language

$$\{a^n b^n \mid n \geq 0\}$$

- ☐  $S \rightarrow aSb$   
☐  $S \rightarrow aS \mid bS \mid \epsilon$   
✓ ☒  $S \rightarrow aSb \mid \epsilon$   
☐  $S \rightarrow aSb \mid ab$

Done